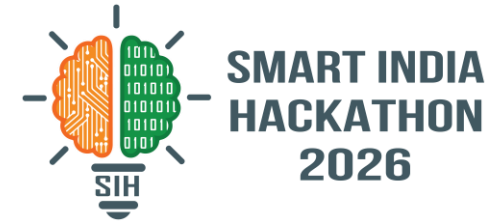


SMART INDIA HACKATHON 2026





IDEA TITLE

THE PROBLEM

- Safety reports (unsafe acts, unsafe conditions, near-misses) contain critical SIF precursor signals
- Manual analysis is slow, inconsistent, and risks missing serious warnings
- HSE teams need a fast, repeatable way to identify which reports signal potential SIF events

THE SAFETYAI SOLUTION

SafetyAI is a working web application that analyzes any free-text safety report and returns structured SIF risk intelligence.



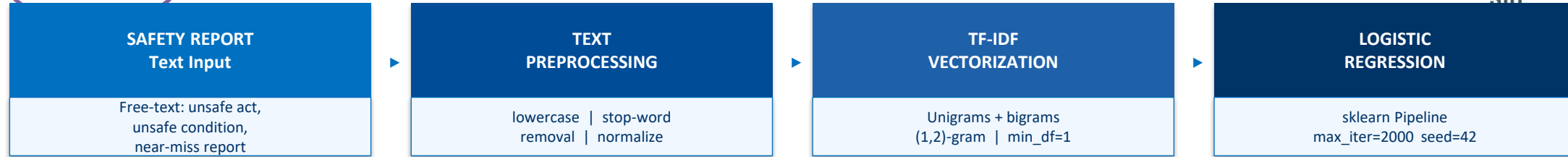
WHAT IT PRODUCES

- ✓ SIF risk level: HIGH / MEDIUM / LOW
- ✓ Model confidence score (0–100%)
- ✓ Detected hazard (9 categories)
- ✓ Detected activity (8 categories)
- ✓ Barrier failure identified (9 types)
- ✓ Matched Life-Saving Rule (9 rules)
- ✓ Evidence-based explanation + action

WHAT IS NOVEL

- Safety Consistency Guard: rule-based override layer
- Transparent ML with feature-weight indicators
- Hybrid: ML prediction + keyword safety enrichment
- Low-confidence flagging (< 50% threshold)
- Full audit trail in SQLite database
- Four-section dashboard web interface

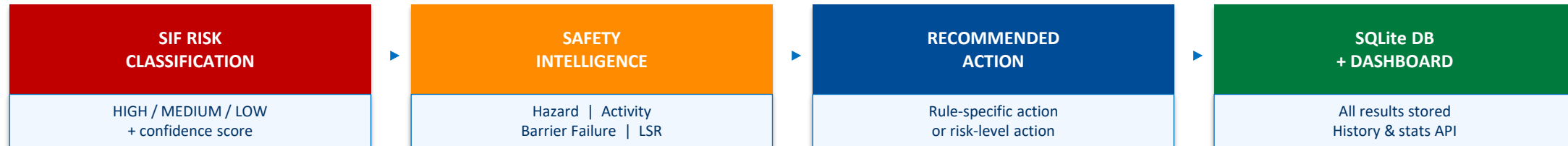
TECHNICAL APPROACH



SAFETY CONSISTENCY GUARD — Rule-based override layer (runs in parallel with ML)

Barrier Failure detected = 3 pts | Critical Hazard = 2 pts | Life-Saving Rule match = 1 pt

Score $\geq 6 \rightarrow$ escalate to HIGH | Score $\geq 3 \rightarrow$ minimum MEDIUM | Score $< 3 \rightarrow$ ML prediction stands



TECH STACK

Python 3 | Flask + Flask-CORS | scikit-learn (TF-IDF + LogisticRegression) | Vanilla HTML / CSS / JavaScript | SQLite

Dataset: safety_reports_expanded_3018.csv (318 original + 2,700 synthetic = 3,018 total reports, balanced across HIGH/MEDIUM/LOW)

FEASIBILITY AND VIABILITY



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2026

WHAT IS IMPLEMENTED

- Working Flask web application
- TF-IDF + Logistic Regression pipeline
- Safety Consistency Guard (rule override)
- 9 Life-Saving Rules (keyword detection)
- 9 Hazard + 9 Barrier + 8 Activity types
- SQLite audit trail and REST API
- Model info + confusion matrix UI
- No paid APIs or cloud required

CHALLENGES

- 2,700 of 3,018 reports are synthetic
- Keyword detection = exact-phrase only
- Phrases outside keyword list = Not detected
- Some predictions are low-confidence (<50%)
- Model not validated on real OIL reports
- Single server; no auth/multi-user support

MITIGATION

- Transparent caveat shown in Model Info tab
- Expand phrase library incrementally
- Add fuzzy / stemmed matching (future)
- Low-confidence flag alerts users to review
- Real OIL data integration as next milestone
- Production deployment & auth are roadmap items

PROTOTYPE EVALUATION EVIDENCE (held-out 20% test split, 604 reports)

- ✓ Accuracy: 98.01% | Precision (macro): 98.04% | Recall (macro): 98.01% | F1-Score (macro): 98.02%
- ✓ HIGH-risk recall: 98.51% | Training set: 2,414 reports | Test set: 604 reports (stratified, seed=42)

⚠ These metrics apply to the prototype dataset (majority synthetic). Performance on unseen real-world OIL reports has not been validated.

IMPACT AND BENEFITS

TARGET USERS: HSE Teams | Safety Officers | Field Operations | Safety Managers

FASTER ANALYSIS

Rapid ML-based review replaces
slow manual reading of reports

EARLY SIF IDENTIFICATION

Surfaces HIGH/MEDIUM-risk reports
before they escalate to incidents

CONSISTENT ASSESSMENT

Repeatable keyword + ML pipeline
removes assessor subjectivity

RISK PRIORITIZATION

Confidence scores + guard override
help teams focus attention

STRUCTURED INTELLIGENCE

Every report yields: Hazard, Activity,
Barrier Failure, Life-Saving Rule

HISTORICAL MONITORING

SQLite audit trail enables trend
analysis across all analyzed reports

ORGANIZATIONAL IMPACT

- Supports proactive, data-driven safety management for HSE teams in oil & gas operations
- Provides a consistent, auditable SIF-precursor assessment process from unstructured report text
- Helps prioritize limited HSE attention toward reports with the most critical safety evidence
- Converts historical safety report data into structured, searchable, actionable safety intelligence

⚠ SafetyAI supports human safety decisions — it does NOT guarantee accident prevention or replace qualified HSE professional judgment.

RESEARCH AND REFERENCES



REFERENCES

- [1] Smart India Hackathon 2026, Problem Statement SIH26165 — Oil India Limited (OIL).
- [2] IOGP Life-Saving Rules framework — referenced directly in model/life_saving_rules.py to define the 9 life-saving rule categories used by SafetyAI.
- [3] SafetyAI dataset — safety_reports_expanded_3018.csv: 318 original reports + 2,700 synthetic reports (generate_synthetic_2700.py), balanced across HIGH/MEDIUM/LOW.
- [4] scikit-learn: Pedregosa et al., JMLR 12, 2011 — TfidfVectorizer + LogisticRegression Pipeline.

FUTURE SCOPE

- | | |
|--|--|
| <ul style="list-style-type: none">→ Real OIL unsafe-act / near-miss reports for training→ Fuzzy / stemmed phrase matching for keyword detection→ Improved handling of ambiguous / low-confidence reports→ Per-user authentication and role-based access | <ul style="list-style-type: none">→ Explainable AI (SHAP/LIME) for transparent predictions→ Automated re-training pipeline as new reports arrive→ Production deployment on OIL's internal safety platform→ Computer vision for safety-image / video analysis (future) |
|--|--|

CURRENTLY IMPLEMENTED: Flask web app + TF-IDF + LR + Safety Guard + SQLite + Dashboard

FUTURE / PLANNED: Real OIL data | Fuzzy NLP | XAI | Auto-retraining | CV

SafetyAI — From Safety Reports to Structured, Actionable Risk Intelligence